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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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Difference-in-differences (DD) is both the most common and the oldest quasi-experimental research design, dating back to Snow’s (1855) analysis of a London cholera outbreak.¹ A DD estimate is the difference between the change in outcomes before and after a treatment (difference one) in a treatment versus control group (difference two): $(\bar{y}_{TREAT}^{POST} - \bar{y}_{TREAT}^{PRE}) - (\bar{y}_{CONTROL}^{POST} - \bar{y}_{CONTROL}^{PRE})$. That simple quantity also equals the estimated coefficient on the interaction of a treatment group dummy and a post-treatment period dummy in the following regression:

$$y_{it} = \gamma + \gamma_i TREAT_i + \gamma_t POST_t + \beta^{2x2} TREAT_i \times POST_t + u_{it} \quad (1)$$

The elegance of DD makes it clear which comparisons generate the estimate, what leads to bias, and how to test the design. The expression in terms of sample means connects the regression to potential outcomes and shows that, under a common trends assumption, a two-group/two-period (2x2) DD identifies the average treatment effect on the treated. All econometrics textbooks and survey articles describe this structure,² and recent methodological extensions build on it.³

Most DD applications diverge from this 2x2 set up though because treatments usually occur at different times.⁴ The processes that generate treatment variables naturally lead to variation in timing. Local governments change policy. Jurisdictions hand down legal rulings. Natural disasters strike across seasons. Firms lay off workers. In this case researchers estimate a regression with dummies for cross-sectional units (α_i) and time periods (α_t), and a treatment dummy (D_{it}):

$$y_{it} = \alpha_i + \alpha_t + \beta^{DD} D_{it} + e_{it} \quad (2)$$

¹ A search from 2012 forward of nber.org, for example, yields 430 results for “difference-in-differences”, 360 for “randomization” AND “experiment” AND “trial”, and 277 for “regression discontinuity” OR “regression kink”.

² This includes, but is not limited to, Angrist and Krueger (1999), Angrist and Pischke (2009), Heckman, Lalonde, and Smith (1999), Meyer (1995), Cameron and Trivedi (2005), Wooldridge (2010).

³ Inverse propensity score reweighting: Abadie (2005), synthetic control: Abadie, Diamond, and Hainmueller (2010), changes-in-changes: Athey and Imbens (2006), quantile treatment effects: Callaway, Li, and Oka (forthcoming).

⁴ Half of the 93 DD papers published in 2014/2015 in 5 general interest or field journals had variation in timing.

In contrast to our substantial understanding of the canonical 2x2 DD model, we know relatively little about the two-way fixed effects DD model when treatment timing varies. We do not know precisely how it compares mean outcomes across groups.⁵ We typically rely on general descriptions of the identifying assumption like “interventions must be as good as random, conditional on time and group fixed effects” (Bertrand, Duflo, and Mullainathan 2004, p. 250), and consequently lack well-defined strategies to test the validity of the DD design with timing. We have limited understanding of the treatment effect parameter that regression DD identifies. Finally, we often cannot evaluate when alternative specifications will work or why they change estimates.⁶

This paper shows that the two-way fixed effects DD estimator in (2) is a weighted average of all possible 2x2 DD estimators that compare timing groups to each other. Some use units treated at a particular time as the treatment group and untreated units as the control group. Some compare units treated at two different times, using the later group as a control before its treatment begins and then the earlier group as a control after its treatment begins. As in any least squares estimator, the weights on the 2x2 DD’s are proportional to group sizes *and* the variance of the treatment dummy within each pair. Treatment variance is highest for groups treated in the middle of the panel and lowest for groups treated at the extremes. This result clarifies the theoretical interpretation and identifying assumptions of the general DD model and creates simple new tools for describing the design and analyzing problems that arise in practice.

By decomposing the DD estimator into its sources of variation (the 2x2 DD’s) and providing an explicit interpretation of the weights in terms of treatment variances, my results

⁵ Imai, Kim, and Wang (2018) note “It is well known that the standard DiD estimator is numerically equivalent to the linear two-way fixed effects regression estimator if there are two time periods and the treatment is administered to some units only in the second time period. Unfortunately, this equivalence result does not generalize to the multi-period DiD design...Nevertheless, researchers often motivate the use of the two-way fixed effects estimator by referring to the DiD design (e.g., Angrist and Pischke, 2009).”

⁶ This often leads to sharp disagreements. See Neumark, Salas, and Wascher (2014) on unit-specific linear trends, Lee and Solon (2011) on weighting and outcome transformations, and Shore-Sheppard (2009) on age time fixed effects.

extend recent research on DD models with heterogeneous effects.⁷ Assuming equal counterfactual trends, Abraham and Sun (2018), Borusyak and Jaravel (2017), and de Chaisemartin and D’Haultfœuille (2018b) show that two-way fixed effects DD yields an average of treatment effects across all groups and times, some of which may have negative weights. My results show how these weights arise from differences in timing and thus treatment variances, facilitating a connection between models of treatment allocation and the interpretation of DD estimates.⁸ I also explain why the negative weights occur: when already-treated units act as controls, *changes* in their treatment effects over time get subtracted from the DD estimate. This negative weighting only arises when treatment effects vary over time, in which case it typically biases regression DD estimates away from the sign of the true treatment effect. This does not imply a failure of the underlying *design*, but it does caution against the use of a single-coefficient two-way fixed effects specification to summarize time-varying effects.

I also show that because regression DD uses group sizes and treatment variances to weight up simple estimates that each rely on common trends between two groups, its identifying assumption is a variance-weighted version of common trends between all groups. The extent to which a group’s differential trend biases the overall estimate equals the difference between how much weight it gets when it acts as the treatment group and how much weight it gets when it acts as the control group. When the earliest and/or latest treated units have low treatment variance, they can get *more* weight as controls than treatments. In designs without untreated units they always

⁷ Early research in this area made specific observations about stylized specifications such as models with no unit fixed effects (Bitler, Gelbach, and Hoynes 2003), or it provided simulation evidence (Meer and West 2013).

⁸ Related results on the weighting of heterogeneous treatment effects does not provide this intuition. Abraham and Sun (2018, p 9) describe the weights in a DD estimate with constant treatment effects as “residual[s] from predicting treatment status, $D_{i,t}$ with unit and time fixed effects.” de Chaisemartin and D’Haultfœuille (2018b, p 7) and Borusyak and Jaravel (2017) describe these same weights as coming from an auxiliary regression and Borusyak and Jaravel (2017, p 10-11) note that “a general characterization of [the weights] does not seem feasible.” Athey and Imbens (2018) also decompose the DD estimator and develop design-based inference methods for this setting. Strezhnev (2018) expresses $\hat{\beta}^{DD}$ as an unweighted average of DD-type terms across pairs of observations and periods.

The challenge in this setting has been to articulate how estimates of equation (2) compare the groups and times depicted in figure 1. We do, however, have clear intuition, for 2x2 designs in which one group's treatment status changes and another's does not. We could form several such designs, estimable by equation (1), in the three-group case. Figure 2 plots them.

Panels A and B show that with only one of the two treatment groups, an estimate from equation (2) reduces to the canonical case comparing a treated to an untreated group:

$$\hat{\beta}_{jU}^{2x2} \equiv \left(\bar{y}_j^{POST(j)} - \bar{y}_j^{PRE(j)} \right) - \left(\bar{y}_U^{POST(j)} - \bar{y}_U^{PRE(j)} \right), \quad j = k, \ell. \quad (3)$$

If instead there were no untreated units, the two way fixed effects estimator would be identified only by the differential treatment timing between groups k and ℓ . For this case, panels C and D plot two clear 2x2 DD's based on sub-periods when only one group's treatment status changes. Before t_ℓ^* , the early units act as the treatment group because their treatment status changes, and later units act as controls during their pre-period. We compare outcomes between the window when treatment status varies, $MID(k, \ell)$, and group k 's pre-period, $PRE(k)$:

$$\hat{\beta}_{k\ell}^{2x2,k} \equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)} \right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)} \right) \quad (4)$$

The opposite situation, shown in panel D, arises after t_k^* when the later group changes treatment status but the early group does not. Later units act as the treatment group, early units act as controls, and we compare average outcomes between the periods $POST(\ell)$ and $MID(k, \ell)$:

$$\hat{\beta}_{k\ell}^{2x2,\ell} \equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)} \right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)} \right) \quad (5)$$

The already-treated units in group k can serve as controls even though they are treated because treatment status does not change.

My central result is that any two-way fixed effects DD estimator is a weighted average of well-understood 2x2 DD estimators, like those plotted in figure 2. To see why, first assume a